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Control and Estimation of Posture During Quiet Stance Depends on Multijoint Coordination

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Hsu W-L, Scholz JP, Schöner G, Jeka JJ, Kiemel T. Control and estimation of posture during quiet stance depends on multijoint coordination. *J Neurophysiol* 97: 3024–3035, 2007. First published February 21, 2007; doi:10.1152/jn.01142.2006. This study tested the hypotheses that all major joints along the longitudinal axis of the body are equally active during quiet standing and that their motions are coordinated to stabilize the spatial positions of the center of mass (CM) and head. Analyses of the effect of joint configuration variance on the stability of the CM and head positions were performed using the uncontrolled manifold (UCM) approach. Subjects stood quietly with arms folded across their chests for three 5-min trials each with and without vision. The UCM analysis revealed that the six joints examined were coordinated such that their combined variance had minimal effect on the CM and head positions. Removing vision led to a structuring of the resulting increased joint variance such that little of the increase affected stability of the CM and head positions. The results reveal a control strategy involving coordinated variations of most major joints to stabilize variables important to postural control during quiet stance.

INTRODUCTION

Upright standing is one of the most common postures by which humans interact with their environment. Successful stabilization of the passively unstable, multisegmented body requires coordinated action of the body's many components (e.g., joints or muscles) to preserve upright posture, particularly while performing other tasks [rapid gesturing with the arms, juggling, etc.; (Balasubramaniam et al. 2000)]. How these many degrees of freedom (DOFs) are organized around the task of postural control is an important question both scientifically and practically, given the high incidence of falls in the elderly population. In addition, it is unclear which variables are used by the CNS to generate the plan for achieving a stable upright posture.

Stabilization of the spatial position of the center of mass (CM) is often assumed to be the goal of postural responses, either implicitly or explicitly (Corriveau et al. 2004; Gage et al. 2004; Pedrocchi et al. 2002; Peterka 2002), implying that a particular value or range of values of the CM are planned for. Physically, the consequence of the line of gravity falling outside of the base of support provides credence for this assumption. On the other hand, the whole body CM is a concept, i.e., there is no way to directly sense its location in space. Stabilization of the CM may be the indirect result of the

control of other variables, for example, the position of the joints relative to their baseline position. Only a few studies have employed experimental methods that test this assumption directly (Krishnamoorthy et al. 2005; Scholz et al. 2007). That the CM may be a controlled variable is consistent, however, with recent evidence that neurophysiological mechanisms exist that encode higher level task variables (Bosco et al. 2003; Poppele et al. 2002).

Knowledge of how the joints and muscles of the body are coordinated for postural stability is important for understanding how the body's position in space is estimated by the nervous system. In general, this is a complex problem; one may ask whether this involves estimation of the position of the head in space, of the CM in space, or of all body segments in space? Different sensors provide information in relation to different parts of the body. Obviously, if there is complex movement of most joints during postural sway, then it is difficult to integrate this information and to characterize the position of the body in space in a simple and unified way. Eliminating visual information, particularly when somatosensory information is also reduced, makes this problem even more challenging (De Nunzio et al. 2005; Krishnamoorthy et al. 2005; Ravaoli et al. 2005).

One hypothesis that, if true, would simplify both the control and estimation problem is that the body acts like an inverted pendulum during quiet standing, stabilized primarily by active control of the ankle joint in combination with passive musculoskeletal properties to maintain alignment of other joints (Winter et al. 1998). If most postural sway takes place around the ankle joint, then the position of the head in space, of the CM in space, or of any other point of the body in space are related to each other trivially. This would simplify the integration of sensory information from multiple sources. This hypothesized "ankle strategy" has served as the basis of many recent postural control models (Horak et al. 1990; Jeka et al. 1998; Kuo 1995; Loram and Lakie 2002; Masani et al. 2006; Maurer et al. 2006; McCollum and Leen 1989; Nashner and McCollum 1985; Nashner et al. 1989; Peterka 2002).

Recent work indicates, however, that inverted pendulum models of quiet stance control represent an oversimplification (Alexandrov et al. 2005; Creath et al. 2005; Horak and MacPherson 1996; Krishnamoorthy et al. 2005; Kuo 1995; Park et al. 2004; Zatsiorsky and Duarte 1999, 2000). For example, Alexandrov et al. (2005) provided data suggesting

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that the ankle, knee, and hip joints are coordinated as different “eigenmodes,” independent combinations of all three joints defined by principal component analysis. Whereas the ankle eigenmode involves predominately ankle motion, motion of the knee and hip make important contributions to the control of upright stance as well. Their approach is consistent with the importance of distinguishing between mechanical and functional DOFs (Li 2006).

Creath et al. (2005) used spectral coherence and co-phase analysis to reveal two modes of coupling between variations of leg and trunk segment angles during quiet stance; in-phase coupling was found <1 Hz, whereas a discontinuous shift to anti-phase coupling occurred above that value. This result suggested the simultaneous co-existence of two “excitable” modes of coupling between legs and trunk even during quiet stance.

Finally, two recent studies support the hypothesis that many joints along the kinematic chain are coordinated to achieve postural stability (Krishnamoorthy et al. 2005; Scholz et al. 2007). Krishnamoorthy et al. (2005) studied joint coordination in subjects who stood as quietly as possible on a narrow beam both with and without vision. Significant motions at most joints besides the ankle and hip were found to be coordinated to minimize disturbance of the CM position. This effect was enhanced when vision was eliminated, most of the increased joint variance being consistent with a control strategy that uses redundant joint combinations to control the CM position (Krishnamoorthy et al. 2005). Results of a recent reanalysis of support surface perturbations (Park et al. 2004) also is consistent with this finding. Differences in the configuration of the joints between pre- and postperturbation phases were largely consistent with a motor equivalent response; i.e., different joint combinations between the phases achieved an equivalent CM position (Scholz et al. 2007).

In light of these results, several hypotheses were addressed in the current study. First, we hypothesized that joints other than the ankle and hip would exhibit significant variability during quiet standing on a normal surface. More importantly we hypothesized that the nervous system would coordinate the joint variations such that most variation has little effect on either the CM or head positions, reflecting instead the use of redundant motor patterns to control these performance variables. Such a finding would mean that the CM could still be a stable reference variable for estimating the body's position in

space. It would also indicate that the postural control problem is a truly multidimensional one involving most major joints along the body's longitudinal axis.

METHODS

Subjects

Seven females and three males between the ages 20 and 45 volunteered to participate in this study. The subjects had no musculoskeletal injuries or neurological disorders that affected their ability to maintain stable upright posture. All subjects gave informed consent according to the procedures approved by the Human Subjects Review Committee of the University of Delaware in compliance with ethical standards laid down by the 1964 Declaration of Helsinki.

Experimental setup

Joint position data were collected at a sampling rate of 120 Hz using a VICON (Oxford Metrics) motion-measurement system composed of six infrared cameras placed in a semicircle at the right side of the subject. The system was tested for reliability and resolution given the small variations of joint angles during quiet stance. For the test, spherical reflective markers were mounted on the 30-cm-long arms and at the axis of a clinical goniometer. The angle between the goniometer arms were changed in 1° increments; this was the maximum resolution of the device. Angle increments were performed in two ranges, $85\text{--}95^\circ$ (arms close together) and $175\text{--}185^\circ$ (arms close to or co-linear). The goniometer was also positioned at different spatial locations corresponding to the position of various joints along the vertical axis of a standing person's body. Angles were computed as described in *Data reduction*. Intra-class correlation coefficients (ICC) were computed in SPSS to test for reliability across measurements. ICC values were all >0.999 , whether computed separately for each joint angle range or across both ranges taken together.

Both a one-way repeated-measure ANOVA and corrected *t*-test for comparing adjacent pairs of angles were used to compare separately differences between angles in each of the two ranges separately. Table 1 presents the angles set on the goniometer, corresponding mean angles computed from the reconstructed marker coordinates in Matlab, and the lower and upper bounds of the 95% confidence intervals (CI). As can be seen, the goniometer and measured values were very similar and adjacent angles were well differentiated (average difference between lower and upper bound of 95% CI for a given goniometer position was $0.00063 \pm 0.00026^\circ$). The differences in angles were highly significant in the ANOVA ($P < 0.000000001$) as well as the *t*-test [all $t(23) > 704.0$, $P < 0.000000001$]. The average joint excursions measured across subjects in the present study are listed in Table 2. These results show that the system resolution and reliability

TABLE 1. Goniometer angle (degrees) set by experimenter, mean measured angle and lower and upper bounds of 95% confidence interval for test of motion analysis resolution

Set Angle	Mean Measured Angle	95% CI Lower Bound	95% CI Upper Bound	Set Angle	Mean Measured Angle	95% CI Lower Bound	95% CI Upper Bound
85	85.5062	85.5055	85.5069	175	175.7962	175.7960	175.7963
86	86.4565	86.4561	86.4569	176	176.8542	176.8540	176.8544
87	87.5642	87.5637	87.5646	177	177.7212	177.7209	177.7214
88	88.5651	88.5646	88.5656	178	178.7811	178.7809	178.7813
89	89.4469	89.4464	89.4473	179	179.6647	179.6645	179.6650
90	90.6008	90.6005	90.6013	180	180.5752	180.5749	180.5754
91	91.6336	91.6334	91.6338	181	181.6834	181.6831	181.6838
92	92.6000	92.5993	92.6000	182	182.5853	182.5849	182.5856
93	93.5535	93.5533	93.5538	183	183.5530	183.5528	183.5531
94	94.5728	94.5725	94.5733	184	184.6422	184.6420	184.6424
95	95.5682	95.5679	95.5685	185	185.6160	185.6158	185.6161

TABLE 2. Mean across subjects of the range of joint angle excursions during quiet stance with the eyes open and closed

Vision Condition	Ankle	Knee	Hip	L ₅ /S ₁	C ₇ /T ₁	AO
Eyes open	3.52 ± 2.55	4.56 ± 3.94	2.87 ± 2.09	4.88 ± 2.97	6.71 ± 2.49	9.95 ± 6.10
Eyes closed	4.04 ± 2.28	4.60 ± 3.47	3.60 ± 2.21	5.05 ± 2.99	6.77 ± 3.36	8.52 ± 4.13

Values are means ± SD. AO, atlanto-occipital.

was sufficient to resolve the joint angle variations measured in this experiment.

For this study, the body was assumed to be bilaterally symmetric and all analyses were limited to the sagittal plane. Spherical reflective markers, 1 cm in diameter, were applied to appropriate joint centers of the right side of the subject's body (ankle joint: immediately inferior to the lateral malleolus; knee joint: lateral femoral condyle; hip joint: greater femoral trochanter; shoulder joint: 2 cm inferior to the lateral aspect of the acromion process of the shoulder). In addition, markers were placed directly anterior to the external auditory meatus, just lateral to the spinous process of the seventh cervical vertebrae, and on the skin over the right pelvis, ~20% of the distance from the greater trochanter to the shoulder and one-third of the distance from the posterior to anterior iliac spines to approximate the L₅/S₁ junction (de Looze et al. 1992).

Experimental procedure

Subjects stood barefoot on a force platform with their feet shoulder-width apart and arms comfortably folded across the chest for each trial. The subject's foot position on both sides was marked to maintain the same foot position for each trial. We attempted to achieve consistent postural alignment at the beginning of each trial by aligning the beam of two fixed laser pointers with small red dots that were placed on the skin overlying the acromion process of the shoulder and over the great trochanter. Subjects were instructed to maintain normal quiet stance throughout each trial, allowing their body to sway naturally; they were not to try to keep their body rigidly aligned or to intentionally oscillate their body during the experiment. They were also told to keep their arms folded and motionless during the trials.

Three 5-min trials each were recorded in each of two conditions for each subject. One set of trials was performed under conditions of normal vision (eyes open, EO). During these trials, subjects were instructed to look straight ahead and focus their attention on a scenic picture that was hung 3-m in front of them. In the second condition, subjects had their eyes closed (EC) during the duration of the trial. After randomly selecting the condition for the first trial for each subject, the condition was presented in alternating order on subsequent trials. Subjects were provided as much rest as they desired between trials to avoid fatigue.

Data reduction

RECONSTRUCTION OF MARKER POSITION. The reflective markers were identified and their three-dimensional (3D) positions from the six camera views were reconstructed off-line using VICON processing software. The coordinates of each reflective marker were then low-pass filtered in Matlab with a 5-Hz cut-off frequency. All signals were filtered with a bi-directional, second-order Butterworth digital filter to eliminate artificial phase shifts in the data induced by filtering. Segment lengths of the body were derived from average marker positions over the first 3 s of the first experimental trial.

JOINT ANGLE CALCULATIONS. The reflective marker coordinates at each data sample were used to calculate joint angles at the ankle, knee, hip, lumbosacral junction, cervical spine (C₇–T₁ junction), and atlanto-occipital joint in the sagittal plane. The atlanto-occipital joint was included based on previous work that revealed substantial motion

of this joint independent of that of the C₇–T₁ articulation (Park et al. 2004; Scholz et al. 2007). The angle between adjacent limb segments was calculated using a link-segment model. Each joint angle is defined as in Fig. 1. The positive angles were defined clockwise; the negative angles were defined counterclockwise. The general formula used for joint angle computation was

$$\text{Angle} = \arccos(\text{dot}(V1, V2)) \quad (1)$$

where V1 was the unit vector representing the proximal segment (from distal to proximal ends of the segment) and V2 was the unit vector representing the distal segment.

CM COMPUTATION. The location of the combined center of mass (d_{CM}) of the n-segment body was estimated at each point in time as the sum of the product of each segment's estimated mass (m_i) and its distance (d_i) from the origin obtained from the marker data, divided by the total mass M . The estimated location of each segment's center of mass along its length that was calculated from the marker data, and their mass contribution to the total body mass was estimated from Winter (1990)

$$d_{CM} = \frac{1}{M} \sum_{i=1}^n m_i d_i \quad (2)$$

In current study, we divided the whole body into foot plus shank, thigh, pelvis, and the trunk plus upper limbs, neck, and head segments for whole body CM calculation.

POWER SPECTRUM DENSITY ANALYSIS: CONTRIBUTIONS TO POSTURAL SWAY. Each subject's mean power spectral density (PSD) for each of six joint angles were calculated and averaged over the

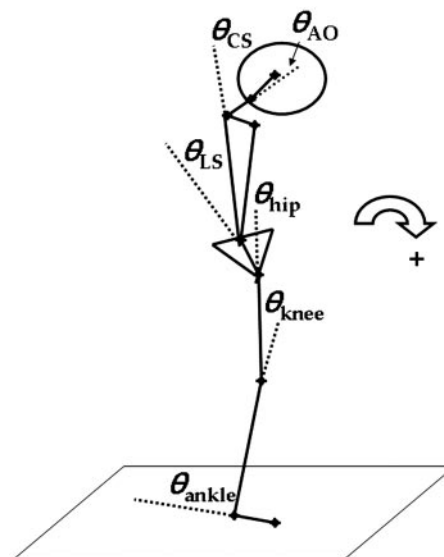


FIG. 1. Schematic illustrations of the angles studied: ankle (θ_{ankle}), knee (θ_{knee}), hip (θ_{hip}), lumbosacral (θ_{LS}), 7th cervical–1st thoracic (θ_{CS}), and atlanto-occipital (θ_{AO}) joints. Positive sign represents clockwise motion.

three 5-min trials of quiet stance in each condition using the Matlab PSD functions that implement Welch's averaging method (Bendat 1980). Calculations used a 20-s Hanning window with a one-half window overlap after subtracting the mean joint angles from the raw data. All PSD functions were analyzed across frequencies from 0.05 to 2.5 Hz in 0.05-Hz increments to avoid the influence of measurement noise at higher frequencies. Spectral density was plotted on a log-log scale to observe the distribution of power in the sway trajectory, which is typically concentrated at low frequencies.

If the body acts primarily as an inverted pendulum during quiet stance, then variance of the ankle should be substantially higher than that of other joints at the dominant frequencies of sway. If a double pendulum model better characterizes the control of quiet stance, then hip and ankle motion should dominate motion of other joints. If instead substantial variation of all joint angles exists, this would be reflected in the spectra and indicate a more complex control structure involving the coordination of movement among most major joints.

JOINT VARIANCE ANALYSIS AND ITS RELATION TO STABILITY OF THE CM POSITION. The question of whether the control system utilizes the available redundancy of the six joint motions to control the CM position during quiet stance on a fixed surface was addressed by using the UCM approach to partition the variability of the joint configuration into two components (Scholz and Schöner 1999). The UCM hypothesis is that the control system selects in the space of the motor elements, here the space of joint postures, a subspace or manifold that corresponds to a value of a task-related variable it is trying to stabilize, e.g., the CM position. In the present context, this UCM subspace is composed of all combinations of joint postures that are equivalent with respect to a given CM position. Therefore variations in joint posture that remain within this subspace have no effect on the CM position or leave it stable. In principle, variations within this subspace can be left uncontrolled, hence, the term "uncontrolled" manifold. We recognize, however, the possibility that other constraints act to limit what portion of the UCM actually is used under different circumstances. In contrast, joint variations lying in a subspace that is orthogonal to the UCM will lead to changes in the CM position away from the mean value with which the UCM is associated. The mean value of the joint configuration is used to estimate the "desired" CM position. Questions about the degree of stabilization of the CM position, then, can be addressed by comparing variances in the two subspaces, i.e., within the UCM (V_{UCM}) and in the orthogonal subspace (V_{ORT}). Given that the analysis is performed with respect to a UCM based on the mean joint configuration, this makes the analysis somewhat conservative. That is, as long as the CM remains well within the limits of the base of support, a range of CM positions are acceptable and consistent with stable upright posture. Thus some amount of V_{ORT} or joint variance leading to different CM positions is expected. A finding that V_{UCM} is substantially larger than V_{ORT} , therefore provides strong evidence in favor of the CM control hypothesis.

During voluntary movements the Jacobian relating changes in the task level variable with changes in the motor elements (i.e., the Jacobian matrix; see following text) varies with the limbs configuration. In that case, the UCM variance analysis is performed at the same point in the trajectory across multiple trial repetitions. In the current experiment on quiet stance, where variations in the joint postures are not induced by voluntary movement and postural variations are presumed to be due to involuntary or random fluctuations, we assume a relatively steady-state posture and perform the analysis across time within each trial (see following text).

Because the amplitude of sway in the anterior-posterior (AP) direction is higher than in the medial-lateral or vertical directions during quiet stance (Winter et al. 1998), we report results of variations within and orthogonal to the UCM in relation to the control of the

anterior-posterior center of mass (CM_{AP}) positions only. This method, then, addressed the extent to which variations in the joint configuration across time led to variations in the CM_{AP} position or tended to preserve the mean CM_{AP} position. Given this partitioning, we can make predictions based on several hypothesized postural control strategies. A value of V_{UCM} found to be significantly higher than V_{ORT} would indicate that variations in the joint configuration tended to lead to a CM_{AP} position consistent with its mean position across time (Fig. 2, B and C). Note that if the postural control system acts primarily as an inverted pendulum (Fig. 2A), most variance is predicted to be V_{ORT} , with little or no V_{UCM} , because movement limited largely to the ankle joint is expected to lead to variations in the CM_{AP} position from moment to moment, consistent with V_{ORT} . Another alternative would occur if the nervous system used a stiffening strategy to stabilize the CM position. Here, one might expect relatively smaller joint variations at all joints considered, i.e., low V_{UCM} and V_{ORT} and $V_{ORT} \approx V_{UCM}$ (Fig. 2D). As discussed in the preceding text, because variations of the CM position normally occur during postural sway (Duarte and Zatsiorsky 1999; Winter et al. 1998), we expect some degree of V_{ORT} to occur that reflects changes in the CM_{AP} position over time. The question is whether there is a substantial and even greater component of variance within the UCM, reflecting more complex coordination of the involved joints to stabilize the average CM_{AP} position.

Details of the mathematical method for performing the UCM analysis has been reported elsewhere (Reisman and Scholz 2003; Scholz and Schöner 1999). The initial step in a formal analysis is to obtain the geometric model relating the task variable (here the CM_{AP} position) to the elemental variables (here the joint angle configuration). In the present experiment, the joint configuration is composed of six angles (angle of the ankle, knee, hip, L_5/S_1 or lumbar spine (LS), C_7/T_1 or cervical spine (CS), and atlanto-occipital (AO) joint angles).

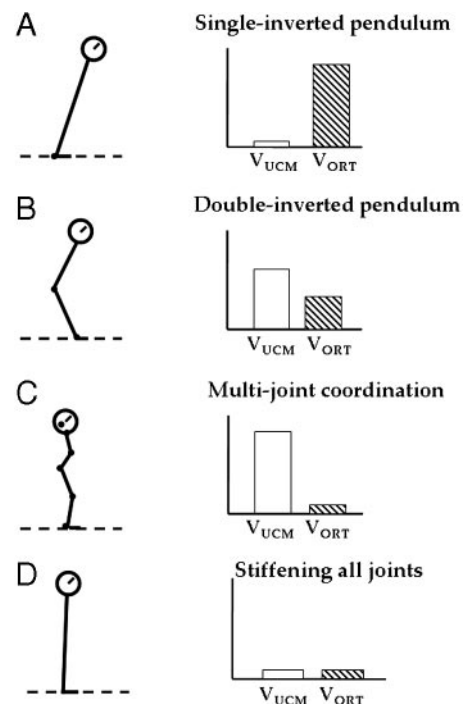


FIG. 2. Illustrations of predicted structure of 2 components of variance, V_{UCM} or joint configuration variance that does not affect the center of mass (CM) position and V_{ORT} , which leads to variability of the CM position, for several hypothetical postural control modes: single-inverted pendulum (A), double-inverted pendulum (B), multi-joint coordination (C), and stiffening strategy (D).

The geometric model relating the CM_{AP} position to the joint configuration, with origin at the ankle is

$$\begin{aligned}
 d_{CM-AP} = & M_{shank} * [d_{shank} * l_{shank} * \cos(\pi - \theta_{ankle})] + M_{thigh} * [l_{shank} * \cos(\pi - \theta_{ankle}) \\
 & + d_{thigh} * l_{thigh} * \cos(\pi - \theta_{ankle} + \theta_{knee})] + M_{pelvis} * [l_{shank} * \cos(\pi - \theta_{ankle}) \\
 & + l_{thigh} * \cos(\pi - \theta_{ankle} + \theta_{knee}) + d_{pelvis} * l_{pelvis} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip})] \\
 & + M_{trunk} * [l_{shank} * \cos(\pi - \theta_{ankle}) + l_{thigh} * \cos(\pi - \theta_{ankle} + \theta_{knee}) + l_{pelvis} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip}) \\
 & + d_{trunk} * l_{trunk} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip} + \theta_{LS})] \\
 & + M_{neck} * [l_{shank} * \cos(\pi - \theta_{ankle}) + l_{thigh} * \cos(\pi - \theta_{ankle} + \theta_{knee}) + l_{pelvis} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip}) \\
 & + l_{trunk} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip} + \theta_{LS}) + d_{neck} * l_{neck} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip} + \theta_{LS} + \theta_{CS})] \\
 & + M_{head} * [l_{shank} * \cos(\pi - \theta_{ankle}) + l_{thigh} * \cos(\pi - \theta_{ankle} + \theta_{knee}) + l_{pelvis} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip}) \\
 & + l_{trunk} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip} + \theta_{LS}) + l_{neck} * \cos(\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip} + \theta_{LS} + \theta_{CS}) \\
 & + d_{head} * l_{head} * (\pi - \theta_{ankle} + \theta_{knee} + \theta_{hip} + \theta_{LS} + \theta_{CS} + \theta_{AO})] \quad (3)
 \end{aligned}$$

where $\theta_{ankle}, \dots, \theta_{AO}$ are the externally defined joint angles; $l_{shank}, \dots, l_{head}$ are the lengths of the respective segments calculated from the static calibration trial; $d_{shank}, \dots, d_{head}$ are the percentages of the segment lengths from the distal end where the mass of that segment lies; and $M_{foot}, M_{shank}, \dots, M_{head}$ are the proportion of total body mass for each of these segments, both estimated from Winter (1990).

Small changes in CM_{AP} are related to changes in joint angles (θ) through the Jacobian, which is the matrix of partial derivatives of the task variable, CM_{AP} with respect to the joint angles (θ).

The second step in the method is to estimate the linear approximation to the UCM from the geometric model. We computed the mean joint configuration across all samples within a trial as corresponding to the hypothetically controlled CM_{AP} position. The mean joint configuration was used to construct the UCM. A linear approximation to the UCM was obtained from the geometric model, linearized around the mean joint configuration

$$CM_{AP} - \overline{CM_{AP}} = J(\bar{\theta}) \cdot (\theta - \bar{\theta}) \quad (4)$$

Here, J is the Jacobian matrix, composed of $\partial CM_{AP} / \partial \theta_i$, where i = [ankle, knee, hip, LS, CS, AO joints]. The linear approximation of the UCM is then the null-space of the Jacobian based on the mean joint configuration (the linear subspace of all deviations from the mean joint configuration that are mapped onto 0 changes in CM_{AP} position by the Jacobian). Matlab was used for the numerical computation of the null space. At each sample value, the deviation of each segment's joint configuration vector from the mean joint configuration vector was obtained. This is essentially a geometric method: the joint deviation vector at each point in time is projected onto the null space, yielding a scalar value that estimates the degree to which that angle configuration is consistent with the mean joint configuration corresponding to a mean CM_{AP} position. The complement of this projection is also obtained providing a scalar estimate of the extent to which this joint configuration leads to a different CM_{AP} position. The computation of V_{UCM} and V_{ORT} was performed on data from 0.5 to 4.5 min, leaving out the first and last 30 s of each 5-min trial. Because the dimensions of the UCM and orthogonal spaces differ, the variances of each subspace were normalized to the number of dimensions of that subspace. Thus the measures of V_{UCM} and V_{ORT} are the variances per DOF. For example, with respect to the hypothesis that the horizontal(AP) CM_{AP} position is stabilized by using flexible combinations of the six joint motions, the task space is one dimensional and the UCM is five dimensional (control of horizontal CM_{AP} hypothesis). Therefore variance of projections within the UCM is divided by 5, whereas variance orthogonal to the UCM is divided by 1.

Although it has been argued that the CM_{AP} is an important control variable for maintaining stable upright posture, the stability of other

variables likely also are important for postural control. Because of the importance of vision and vestibular information (for more dynamic tasks) to postural control, stabilization of the head's position or orientation might also be important. Moreover, because the head is relatively massive, control of CM_{AP} is likely linked to control of the head. Other control hypotheses are also possible. Indeed, it is possible that variance is structured with respect to a variety of variables. Therefore to better assess the extent of CM_{AP} control, we also performed the UCM analysis relative to control of both the head's position and spatial orientation (i.e., with the vertical). In addition, if the body was controlled as a double-inverted pendulum, using combined ankle and hip strategies (Bardy et al. 1999; Creath et al. 2005; Horak and MacPherson 1996; Kuo 1995; Park et al. 2004), then negative covariation of the ankle and hip joints could also lead to CM_{AP} position stabilization. A strategy that limits motion to the hip and ankle (2 DOF) would provide a minimally redundant system for stabilizing the CM_{AP} position (1 DOF). Thus we performed the UCM analysis on a reduced model of CM_{AP} control involving only the ankle and hip joints. The test of this model assumes, therefore, that most motion is restricted to these 2 DOF and that control of upright posture can be assumed to be related to control of these two angles. In this model, the combined mass of the head, arms, and trunk and their resultant location were used to represent the "trunk" segment, whereas the combined mass of the shanks and thighs and their resultant location were used to represent the "leg" segment (Winter 1990). Note that the masses of both legs and both arms were used in all models estimating CM position to get the accurate position along the anterior-posterior and vertical dimensions although this report is limited to the former dimension.

To compare the UCM effect for the different control hypotheses, (i.e., CM_{AP} position based on the 6-DOF model, head position, head orientation and CM_{AP} position based on the 2-DOF model), the ratio of V_{UCM} to V_{ORT} for each control hypothesis was computed for each trial of a given individual's data

$$UCM \text{ ratio} = \frac{V_{UCM}}{V_{ORT}} \quad (5)$$

This ratio then was transformed to natural log value to correct for a nonnormal distribution. Using the ratio is particularly important for comparing the 6- and 2-DOF model of CM_{AP} control because of the different number of total joints contributing to the variance of each hypothesis, whereas the hypotheses about controlling the head position and spatial orientation are also based on 6 DOF.

Statistics

Paired t -test were performed to compare the total joint configuration variance as well as the actual variance of the CM_{AP} across the two vision conditions. A two-way repeated-measures ANOVA (variance component by vision condition) was performed to determine how the variance of joint motion was organized and its dependence on the vision condition for each control hypotheses. When there was a significant interaction related to our hypotheses, planned contrasts were performed using the m-matrix structure in SPSS.

A two-way repeated-measures ANOVA was performed to compare the different control hypotheses (CM_{AP} and head positions and head orientation, based on 6-DOF model, and CM_{AP} position based on 2-DOF model) and their relationship to vision condition, using the log transform of the ratios of V_{UCM} to V_{ORT} . A significant interaction related to our hypotheses was examined using the m-matrix structure in SPSS.

The log(PSD) value for joint angles was analyzed using a three-way repeated-measures ANOVA with vision (2), joint (6) angles and frequencies (50) as factors. When there was a significant interaction effect of joint angles $\times$ frequencies, paired t -test at each of the 50 frequencies were performed to determine the frequency range of the

differences. To control the false discovery rate from the multiple testing, the method of Benjamini and Hochberg (1995) was applied to the resulting P values. All statistical analyses were performed using the SPSS 13.0 package and the α -level was set at 0.05. Cross-spectral analysis of all possible joint pairs was also performed and the peak of the coherence function examined to determine if covariation of individual pairs of joints could explain any observed UCM effects.

RESULTS

Overall joint variability

The actual range of motion at each joint in the static posture was quite small. The *top six panels* of Fig. 3 show the fluctuations of each measured joint angle across all three trials for a given exemplar subject (*subject 3*), for both EO condition (A) and EC condition (B). The *bottom panel* of the figure shows the variations of the CM_{AP} position for the same trials. Figure 4 shows comparable data for one atypical subject (*subject 10*) who showed a more noticeable slow drift of the joint angles across the 5-min trials superimposed with the faster fluctuations. Note that in both subjects, the CM_{AP} position appears to be relatively stable across time compared with the joint angles, suggesting that many different combinations of joints were used in an attempt to attain a stable CM_{AP} position. Another way of looking at this is that joints compensated for each others fluctuations to achieve a more stable CM_{AP} position, which would require interjoint coordination.

Note, however, that such qualitative comparisons between joint variations and CM_{AP} variations are problematic because of differences in the units of measurement (i.e., radian vs. meter) and dimensions (6 vs. 1) for these two types of variables. However, the UCM approach provides a means to quantitatively make this comparison by formulating questions

about control of task-related variables such as CM position in the space of the motor elements, here the joint angles. If the CM_{AP} position is kept more stable than the joint motions themselves, then significantly more joint configuration variance should lie in the UCM instead of the complementary subspace of joint space.

Table 3 presents the variance across all samples within a given trial of joint excursion and averaged across three trials for a given subject. As might be expected, the mean variance for each joint as well as the total joint configuration variance was higher in the EC condition than in the EO condition. The total joint configuration variance (i.e., geometric mean of individual joint variances) was 0.0019 ± 0.002 radian²/DOF in EO condition and 0.0029 ± 0.003 radian²/DOF in EC condition. The paired t -test showed a significant difference in total joint variance between EO and EC condition ($t = -3.01$ $P < 0.05$). Note that the proximal joint angles exhibited larger variance than did the distal joint angles.

Variance of CM position

As noted in the preceding text, the *bottom panel* of Figs. 3 and 4 illustrates the fluctuations of the CM_{AP} position for all three trials for a typical subject (*subject 3*) and the atypical subject who exhibited a slow drift of the mean joint posture of his joints across the trial (*subject 10*). Table 4 presents the variance of the CM_{AP} position across all samples within a trial, and then averaged across three trials for a given subject. Note that the SE was very small for both vision conditions. The variance of the CM_{AP} position was generally larger in the EC condition than in the EO condition. However, this difference did not reach statistical significance ($t = -1.06$, $P = 0.317$), indicating that the differences in joint variance between vision

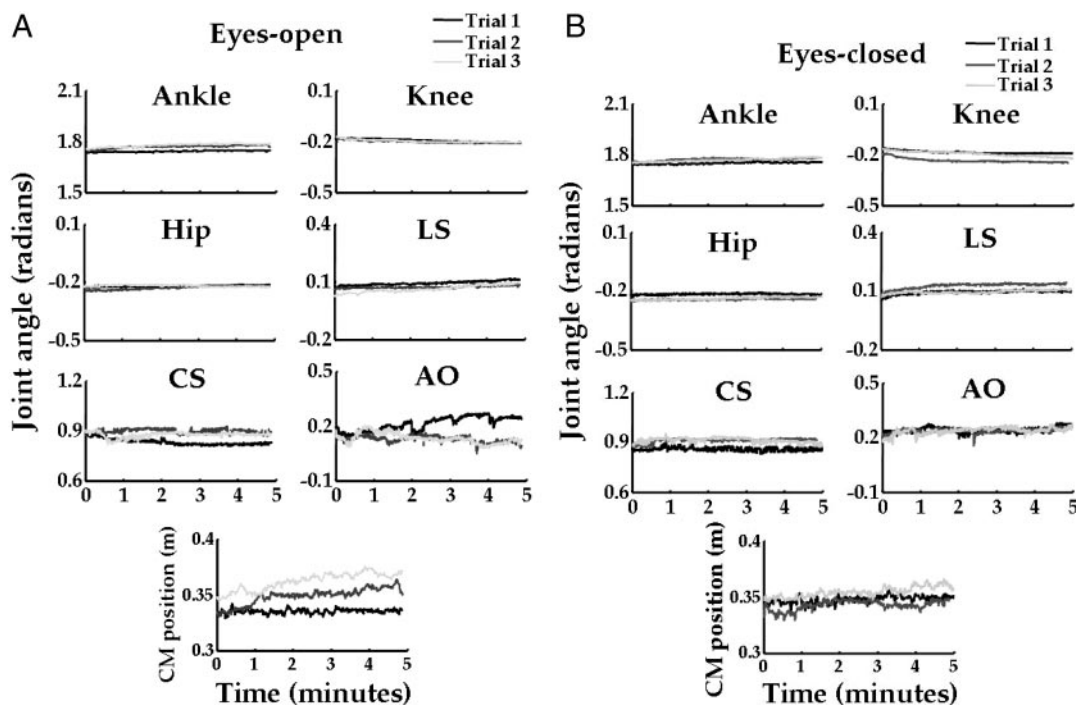


FIG. 3. Joint excursions at ankle, knee, hip, lumbosacral (LS), C₇–T₁ or cervical spine (CS) and atlanto-occipital (AO) joint and the CM trajectory with respect to origin of the coordinate system (*bottom*) for each trial of an exemplar subject (*subject 3*) in the eyes-open (EO) condition (A) and eyes-closed (EC) condition (B). Each line represents a single trial.

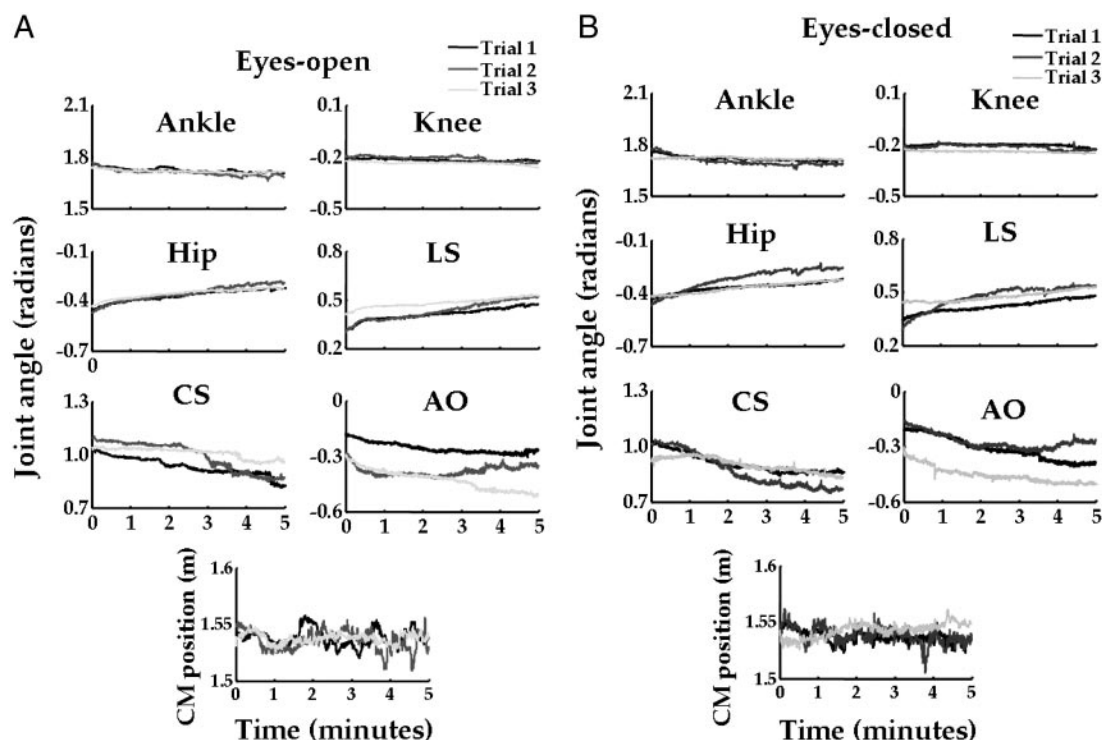


FIG. 4. Joint excursions at ankle, knee, hip, lumbarsacral (LS), C_7 – T_1 or CS and AO joint and the CM trajectory with respect to origin of the coordinate system (bottom) for each trial of an atypical subject (subject 10) who had a gradually drift along the time in the EO condition (A) and EC condition (B). Each line represents a single trial.

conditions did not translate directly into variance of the CM_{AP} position.

Frequency analysis of joint motion

Another way to address differences in variance among different joints and the effect of vision is to compute the PSD function of each joint angle. This allowed us to determine if the differences in overall joint variance depended on the sway frequency. The results for the mean PSD of each joint angle across all subjects in both EO and EC conditions are presented in Fig. 5, A and B, respectively. Main effects for vision, joints, and frequency were found in the repeated-measures ANOVA [vision: $F(1,9) = 5.4$, $P < 0.05$; joint: $F(5,45) = 51.4$, $P < 0.001$; frequency: $F(49, 441) = 666.23$, $P < 0.001$]. Spectral power was larger in the EC than in the EO condition, consistent with the direct analysis of joint variance. The AO joint showed higher power than any other joint at all spectral frequencies. Of particular interest is that the ankle and hip joint powers tended to be lower than those of several other joints, particularly at the lower frequencies where most of the power lies (Fig. 5, A and B).

The results of spectral analyses suggest that all joint motions need to be coordinated to achieve postural stability. We asked here whether this coordination is reflected by a strong coupling of individual pairs of joint motion or is the result of a more complicated coordination strategy, given the observed redundancy of joint motions. Thus we analyzed coherence functions of cross-spectral analysis of all pairs of the six joints examined. The results are presented in Table 5 for all pairs of joints. The results show that the coherence between pairs of joint angles was found to be quite low, typically < 0.5 . Only a few adjacent

joints had coherence of joint motion > 0.5 . Thus although joint variance was apparently organized to limit CM_{AP} variance, as suggested by the significant effect of vision on joint variance and nonsignificant effect on the CM_{AP} position, this was apparently not achieved by a straightforward, consistent covariation of particular joint pairs but involved a more complicated coordination of the joints. To better understand the relationship between the joint coordination and its relationship to control of the CM_{AP} position as well as other putative control variables, we next performed a UCM analysis of joint variance which examines the structure of this variance with respect to the hypothesized controlled variables.

Structure of joint configuration variance related to control of the CM_{AP} position

Given that there is relatively equal variability of all measured joints during quiet upright stance, a complex form of coordination among the joints likely accounts for the relatively stable CM_{AP} position depicted in Figs. 3 and 4. The UCM analysis allowed us to link joint angle variance to variations of the CM_{AP} position via a geometric model. Figure 6A presents the mean ($+1$ SE of estimation) of V_{UCM} and V_{ORT} related to control of the CM position based on the 6-DOF model.

Variance representing joint configurations that were equivalent with respect to producing the mean CM_{AP} position (i.e., V_{UCM}) was significantly higher than joint variance that leads to a different CM_{AP} position [$F(1,9) = 10.6$, $P < 0.01$]. A significant interaction of vision condition by variance component was also found [$F(1,9) = 7.6$, $P < 0.05$]. This was due to higher V_{UCM} for the EC condition ($F_{1,9} = 7.8$, $P < 0.05$; Fig. 6A), while V_{ORT} did not differ between the EO and EC

TABLE 3. Variance of joint excursion

Subject ID	Ankle			Knee			Hip			L ₅ -S ₁			C ₇ -T ₁			AO		
	EO	EC	EO	EO	EC	EO	EO	EC	EO	EO	EC	EO	EO	EC	EO	EO	EC	EO
1	0.104	0.221	0.224	0.224	0.326	0.029	0.029	0.119	0.313	0.496	0.188	0.682	0.720	1.093				
2	0.019	0.140	0.032	0.032	0.127	0.012	0.012	0.047	0.055	0.144	0.183	2.001	0.264	0.496				
3	0.064	0.059	0.067	0.067	0.146	0.038	0.038	0.021	0.151	0.112	0.240	0.153	0.835	0.219				
4	0.141	0.339	0.256	0.256	0.387	0.142	0.142	0.024	0.439	0.242	0.495	0.429	0.475	1.990				
5	0.150	0.153	0.713	0.713	0.854	0.160	0.160	0.103	0.588	1.078	0.419	0.563	0.423	0.574				
6	0.025	0.023	0.033	0.033	0.026	0.022	0.022	0.018	0.109	0.115	0.139	0.305	0.154	0.454				
7	0.034	0.031	0.012	0.012	0.054	0.051	0.051	0.054	0.169	0.154	0.099	0.322	0.449	1.794				
8	0.023	0.039	0.072	0.072	0.038	0.141	0.141	0.036	0.070	0.085	0.244	0.242	0.095	0.131				
9	0.133	0.220	0.494	0.494	0.638	0.066	0.066	0.071	0.065	0.279	0.140	0.095	0.352	0.082				
10	0.198	0.300	0.105	0.105	0.091	1.409	1.409	1.697	1.671	1.849	3.283	4.188	1.454	2.529				
Mean	0.089 ± 0.020	0.153 ± 0.036	0.201 ± 0.074	0.201 ± 0.074	0.269 ± 0.090	0.207 ± 0.135	0.207 ± 0.135	0.219 ± 0.165	0.363 ± 0.156	0.456 ± 0.182	0.543 ± 0.307	0.898 ± 0.405	0.522 ± 0.126	0.936 ± 0.276				

Units are radian² × 10⁻³. Means ± SE are shown.

TABLE 4. Variance of CM position excursion

Subject ID	EO	EC
1	0.072	0.096
2	0.018	0.029
3	0.038	0.016
4	0.023	0.044
5	0.032	0.041
6	0.025	0.017
7	0.020	0.028
8	0.019	0.035
9	0.036	0.032
10	0.049	0.043
Mean	0.033 ± 0.005	0.038 ± 0.007

Units are m² × 10⁻³.

conditions ($P = 0.301$). For the increase in joint variance to be restricted largely to the subspace in joint space that does not affect the CM_{AP} position when removing vision requires more complicated coordination of all involved joints that is apparently not reflected in the coherence between the motion of individual joint pairs.

Structure of joint configuration variance related to the stability of other hypothesized control variables

The structure of joint variance, when considered in relation to control of head position (Fig. 6C) and head orientation (Fig. 6D), was similar to that when considering CM_{AP} control involving the 6-DOF model. V_{UCM} was significantly higher than V_{ORT} with respect to stabilizing both the head's position [$F(1,9) = 11.4$, $P < 0.01$; Fig. 6C] and its spatial orientation [$F(1,9) = 8.1$, $P < 0.05$; Fig. 6D]. There was also a significant effect of vision, with joint configuration variance being higher for the EC condition [head position: $F(1,9) = 7.7$, $P < 0.05$; head orientation: $F(1,9) = 10.9$, $P < 0.01$]. An interaction between vision condition and variance component was present, however, only in relation to control of the head's spatial position [$F(1,9) = 7.7$, $P < 0.05$]. As with CM_{AP} control, V_{UCM} was significantly higher for the EC condition [$F(1,9) = 7.7$, $P < 0.05$; Fig. 6A], whereas V_{ORT} did not differ between the vision conditions ($P = 0.345$). Regarding head orientation, removal of vision resulted in a relatively proportional increase in both V_{UCM} and V_{ORT} .

If the body was to behave as a double-inverted pendulum, with most joint fluctuations occurring about the hip and ankle joints, coordination of the two joints could achieve a similar stabilization of the CM across time by negative covariation of the two joints. This effect might be limited, however, because this model of postural control is minimally redundant. Indeed when the UCM analysis was performed with respect to the two-joint model, a different result occurred (Fig. 6B). There were no significant differences between V_{UCM} and V_{ORT} ($P = 0.818$), no effect of the vision condition ($P = 0.385$), and no interaction between these factors ($P = 0.386$). Thus an analysis based on assuming that the body acts as a double-inverted pendulum, anticipated by the joint spectral analysis results, would lead to the conclusion that the CM_{AP} position is not a well-stabilized variable of the control system. Based on the joint variance and spectral analysis results and the UCM analysis based on the full six-joint model, we concluded that this assumption is not correct.

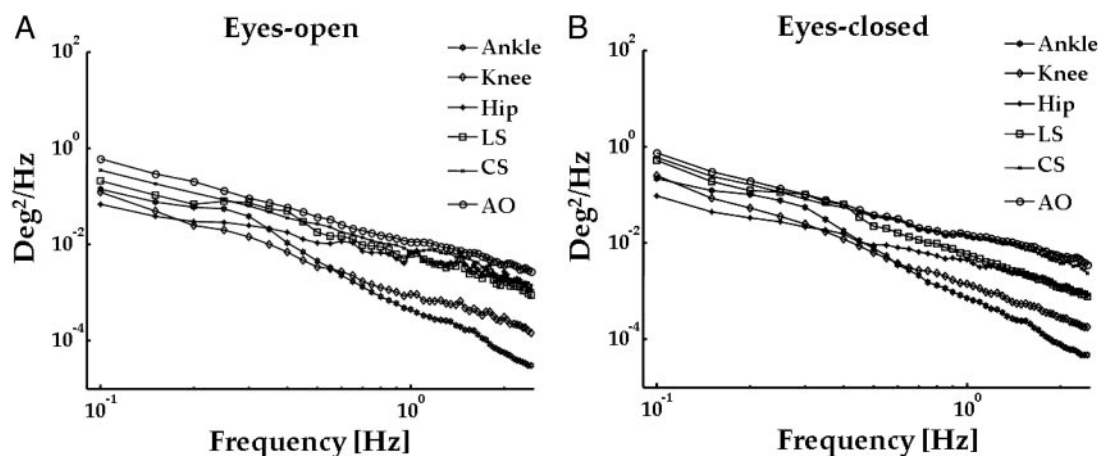


FIG. 5. Mean of power spectral measures of joint sway angle for ankle, knee, hip, lumbosacral (LS), C₇–T₁ or cervical spine (C₇) and AO joint across subjects in EO condition (A) and EC condition (B).

Comparison of various control hypotheses

Because of the different number of DOF included in the 6-DOF model of CM_{AP} control compared with the 2-DOF model, we decided to directly compare the different control hypotheses by comparing the ratio of V_{UCM}/V_{ORT} , reflecting the relative amount of variance per DOF in each subspace, rather than individual variance components. The ratio was computed separately from the results of the UCM analysis performed on each trial. Then each subject's median value across trials was log transformed to account for the nonnormal distribution of this variable and subjected to a repeated-measure ANOVA to compare the different control hypotheses. The results are presented in Fig. 7.

Comparing V_{UCM}/V_{ORT} across all four control hypotheses revealed a significant difference [$F(3,27) = 25.9, P < 0.001$]. The ratio of variance components related to control of the CM_{AP} position based on the 6-DOF model was substantially larger than the ratio related to control of head orientation [$F(1,9) = 74.0, P < 0.001$] or related to control of the CM_{AP} position that was based on a simplified two-joint model [$F(1,9) = 39.7, P < 0.001$]. There was no difference, however, between the ratios related to control hypotheses based on the 6-DOF CM_{AP} position and head position ($P = 0.267$, Fig. 7). The variance ratio related to head position control also was

significantly larger than that related to control of head orientation [$F(1,9) = 83.52, P < 0.001$]. Both the overall effect of vision ($P = 0.762$) and interactions of vision condition with different control hypotheses were also not significant ($P > 0.1$).

DISCUSSION

The study of upright posture has been strongly influenced by the inverted pendulum hypothesis. According to this hypothesis, postural control largely involves control of the ankle joint, in combination with passive musculoskeletal properties to maintain alignment of other joints (Jeka et al. 1998; Kuo 1995; Loram and Lakie 2002; Masani et al. 2006; Maurer et al. 2006; McCollum and Leen 1989; Nashner and McCollum 1985; Nashner et al. 1989; Peterka 2002; Winter et al. 1998). This strategy would establish a direct correspondence between motion at the ankle joint and the body's position in space. As a result, both postural control and estimation of the position of the body in space would be vastly simplified. Although attrac-

TABLE 5. Maximum coherence of joint sway angle between pairs of joints

Joint Pair	EO	EC
Ankle-knee	0.50 ± 0.05	0.63 ± 0.05
Ankle-hip	0.31 ± 0.07	0.36 ± 0.06
Ankle-LS	0.35 ± 0.06	0.43 ± 0.07
Ankle-CS	0.23 ± 0.06	0.28 ± 0.07
Ankle-AO	0.11 ± 0.02	0.23 ± 0.06
Knee-hip	0.45 ± 0.07	0.39 ± 0.06
Knee-LS	0.39 ± 0.06	0.45 ± 0.06
Knee-CS	0.26 ± 0.06	0.24 ± 0.04
Knee-AO	0.11 ± 0.03	0.17 ± 0.04
Hip-LS	0.55 ± 0.09	0.51 ± 0.09
Hip-CS	0.33 ± 0.08	0.27 ± 0.07
Hip-AO	0.19 ± 0.03	0.17 ± 0.05
LS-CS	0.32 ± 0.07	0.36 ± 0.06
LS-AO	0.17 ± 0.04	0.30 ± 0.05
CS-AO	0.45 ± 0.05	0.48 ± 0.08

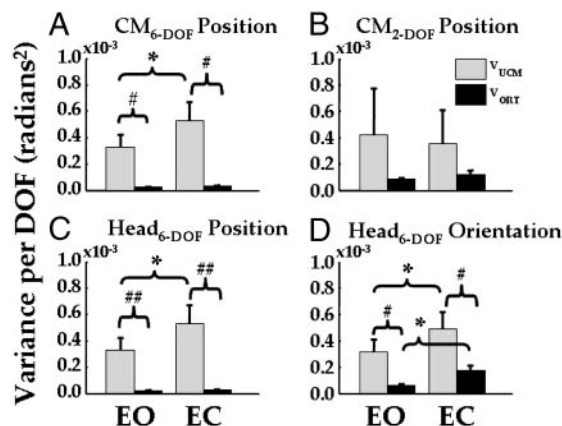


FIG. 6. Mean $\pm$ SE of joint configuration variability per degree of freedom (DOF) underlying control of the CM_{AP} position based on a 6-DOF model (A) and based on a 2-DOF, ankle-hip model (B), and control of the AP position of the head (C) and head's orientation in the sagittal plane (D), both based on a 6-DOF model. Adjacent pairs of bars represent V_{UCM} and V_{ORT} for EO condition (left) and EC condition (right). #, significant differences between V_{UCM} and V_{ORT} ; *, significant differences between EO and EC; ## and **, $P < 0.01$; # and *, $P < 0.05$.

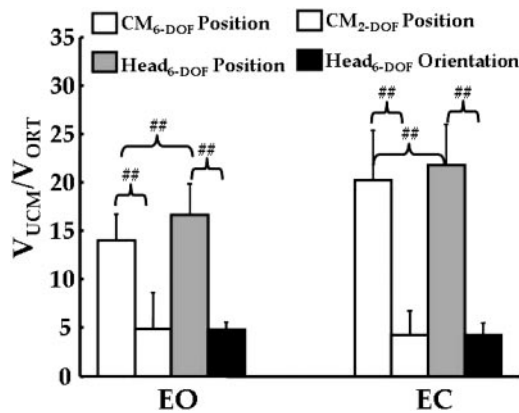


FIG. 7. Ratio of V_{UCM} to V_{ORT} for each control hypothesis described in Fig. 6. Left: EO; right: EC. Symbols are the same as in Fig. 6.

tive, this strong version of the inverted pendulum hypothesis has been challenged based on several lines of evidence. For example, Alexandrov et al. (2005) showed that the ankle, knee, and hip joints are coordinated into different modes of action for various tasks, including postural control, whereas Creath et al. (2005) revealed signatures of both in- and anti-phase coupling of the legs and trunk during quiet stance.

Most major joints along the body's longitudinal axis are coordinated to stabilize upright posture.

Spectral analysis performed on all six measured joint angles in the current study revealed considerable variance at the relevant frequencies of postural sway in all cases. The ankle and hip joints were not found to be special among the joint DOFs examined. In terms of overall variability, these joints ranked fourth and fifth with more variance present at the atlanto-occipital and LS joints. In addition, the computed coherence between pairs of joint angles was found to be quite low, in most cases well <0.5 . This underscores that correlations between pairs of joint angles is too weak for motion of one joint to represent motion at the other. Taken together, these observations run counter to a strict ankle or even ankle-hip control strategy for postural control.

The strong UCM effect found in relation to control of the spatial positions of the CM and head provide support for a more complex control scheme that involves most joints along the body's longitudinal axis. Only a small portion of the measured temporal variability of the joint configuration affected either the CM or head positions. If most motion was to take place at the ankle joint, as would be the case if the body acted as an inverted pendulum, then ankle joint variability would load exclusively on V_{ORT} , leading to variable CM and head positions (Fig. 2A), contrary to our results. Moreover, the UCM effects obtained with a two joint ankle-hip model (Fig. 2B) did not approximate the effects of the more realistic six-joint model (Fig. 6A). In fact, test of the reduced model resulted in no significant difference between the two variance components (Fig. 6B), although there was a tendency for V_{UCM} to be larger than V_{ORT} (Fig. 6B). In addition, this result was inconsistent with the measured changes in the variance of CM_{AP} ; the 2-DOF result predicts a substantial increase in CM variance in the EO condition due to higher V_{ORT} .

Observed UCM structure of variance is consistent with evidence from other recent studies

An earlier study by Krishnamoorthy et al. (2005) analyzed joint variance in participants who stood on a narrow support surface. The results revealed the same structure of variance observed in the present study. All measured kinematic DOFs were engaged, but more variance was generated in directions in joint space that left the CM position invariant. DOFs were thus coordinated to stabilize the CM position. When the sensory conditions were made more difficult by removing vision, there were significant increases in joint variance. These increases had a limited effect on the CM position, however, being primarily directed into the UCM. In other words, combinations of joint motions leading to an equivalent CM position were more extensively activated than combinations that would shift the CM position (Krishnamoorthy et al. 2005; Scholz et al. 2001). The observation of the same UCM structure of variance in the present study confirms that this effect is not a consequence only of the limited postural support on a narrow support surface. Instead, the UCM structure of variance can be readily observed when standing on a normal surface as well.

How does the UCM structure of variance relate to postural adjustments after perturbations? Unlike the typically emphasized ankle and hip strategy, recent work suggests that both response components are present, their relative contributions depending on the nature of the perturbation and the stance conditions (Alexandrov et al. 2005; Park et al. 2004). A re-analysis of the data of Park and colleagues (2004) decomposed postural adjustments induced by perturbations of the support surface into joint configuration changes that leave the CM unchanged (changes within the UCM) and joint configuration changes that shift the CM (changes orthogonal to the UCM) (Scholz et al. 2007). The analysis found that significantly more of the postural adjustment occurred within the UCM than orthogonal to it. Given that postural adjustments at the ankle load primarily on the subspace orthogonal to the UCM (causing CM motion), this means that the postural response was not primarily ankle motion or even a combination of ankle and hip motion (Scholz et al. 2007) but involved substantial motion at all measured joints. This response pattern was particularly prominent for responses to low-amplitude perturbations.

Based on spectral coherence and co-phase analyses, Creath et al. (2005) reported two modes of coupling between the trunk and the leg segment of the body. Coherent in-phase coupling occurred at frequencies <1 Hz. At ~ 1 Hz, they described a discontinuous shift to anti-phase coupling between these two body segment angles, although the anti-phase pattern was associated with far less spectral power than the in-phase pattern. Based on these results, the authors proposed that the control structure underlying quiet stance involves two excitable modes of coupling that are always present but differentially scaled depending on environmental and task conditions or on the nature of perturbations. The present results expand this view and suggest an even more complex control structure involving the variation of most major joints. Thus while there is coherent movement of the body in space, illustrated by coherent in-phase movement between leg and trunk segmental angles, this movement is superimposed on low coherence between pairs of joint angles (Table 5). We should point out

that we replicated with the current data set (not reported) the co-phase and coherence analyses of Creath et al. (2005), confirming that our data were obtained under similar conditions.

A recent study by Freitas and colleagues used the UCM approach to study postural responses to voluntary shifts in the COP (Freitas et al. 2006). Subjects achieved these shifts by oscillating either about the ankle, knee, or hip (which included trunk flexion-extension). Both the instantaneous CM position and the trunk's orientation in space (assessed across similar cycles) were found to be stabilized by coordinated changes in joint motions. However, the UCM effects ($V_{\text{UCM}} > V_{\text{ORT}}$) were relatively weak with respect to CM control compared with stabilization of trunk orientation. We did not examine trunk orientation here (Scholz et al. 2007), but the different results concerning CM control are likely due to several factors. First, their geometric model included only the hip, knee, and ankle joints. As was shown in the current experiments, the 2-DOF model of CM control produced relatively weak UCM effects. Second, the two tasks differed substantially. Although we studied the postural steady state, their results come from attempts to voluntarily change the postural state; this may partially account for the relatively weak effects on CM stabilization.

Effect of vision on joint coordination and implications for proprioception

When vision was eliminated by asking subjects to close their eyes (EC), overall joint configuration variance increased compared with the eyes open (EO) condition. However, this increase was not general as one might expect if the loss of vision were to lead to an overall upregulation of proprioceptive gains, accompanied by some increase in noise levels. Instead, the increased joint configuration variance was found to lie mostly in the UCM versus the region of joint space that would cause increased CM or head position variability (Fig. 6, A and C). This result contrasts with a relatively equal increase in both variance components when considering the control of head orientation (Fig. 6D) or no effect when considering the 2-DOF CM_{AP} control model (Fig. 6B). The fact that head orientation showed such an increase in the component orthogonal to its UCM indicates that the opposite observation regarding control of the CM and head position based on the 6-DOF model is not trivial.

Examining posture on narrow support, Krishnamoorthy et al. (2005) recently provided a hypothesis to explain this result. They reported a similar but even stronger differential effect of removing vision on the two variance components. In view of the substantial amount of variance observed in all measured joints and the UCM structure found in that variance ($V_{\text{UCM}} \gg V_{\text{ORT}}$), Krishnamoorthy et al. hypothesized that the problem that the postural system needed to solve on the narrow base of support was to keep the body scheme updated so as to be able to predict which joint motions are allowed and which are not. That requires estimates of all joint angles and joint velocities. Because proprioception depends on the dynamic contribution of muscle receptors (Clark et al. 1989), increasing joint variability may be a useful strategy to increase the sensory signal from proprioceptive systems that support estimation of the whole body joint configuration. By allowing an increase in

joint motion to occur primarily within the UCM, the negative impact of such a strategy on variance of the critical variables CM and head position is limited.

More generally, proprioception and the estimation of the whole body joint configuration play a much more important role in the UCM based conception of postural control than, for example, within the inverted pendulum conception. In the UCM-based conception, the sizeable variance at all measured joints needs to be coordinated such that postural sway of the CM and head position is minimized. This requires an estimate of all joint angles, that is, a representation of the body scheme. A direct test of this consequence of the UCM picture of postural control could be based on the prediction that patients with a complete loss of proprioception should not show the same variance structure as shown here.

Revised conception of postural control

This revised argument for the importance of CM and head position emphasizes the importance of relating sensory information across different reference frames. Because most joints along the body axis are highly variable, these transformations are not trivial and are all the more critical. The observed UCM structure in joint space simplifies some of these transformations but conversely relies on reliable estimates of the body configuration. Maurer and colleagues recently proposed a multi-sensory model of control of upright stance that might help simplify the problem of integrating sensory information (Maurer et al. 2006). The model involves the up- and down-streaming of sensory information (ankle proprioceptors, pressure sensors at the feet, and vestibular receptors). In model simulations that combined sensor fusion and sensory thresholds, an automatic context-specific sensory re-weighting of these different sources occurred across different stimulus conditions. Our results highlight the need for an understanding of the transformations modeled by Maurer et al. Their model was, however, based on a single-inverted pendulum model of the body. Accounting for the coordination among joints that underlies the UCM structure of variance will complicate the picture but should provide important insights into how sensory-motor integration is achieved in the postural system.

In conclusion, the observation of a UCM pattern of joint variance both confirms and disputes common assumptions about the nature of the postural system. The UCM result disconfirms that the CNS simplifies the postural control problem by using primarily a single DOF, the ankle, to achieve stability of the body in space, and is consistent with other recent results (Alexandrov et al. 2005; Creath et al. 2005). The UCM result confirms, however, that the CNS makes special efforts to limit sway of the body by coordinating variance at all joints such that most joint motion is decoupled from motion of the body in space. This finding has important implications for problems of both estimation and control.

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